

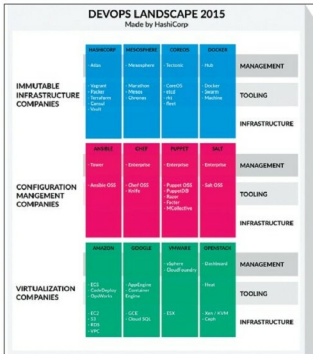


Figure 3: The DevOps landscape 2015 by HashiCorp

create Atlas. CoreOS recently released Tectonic to bundle *etcd*, *rkt* and *flannel* into a management layer. The infographic in Figure 3 shows the tooling portfolios of the major players in these categories and how those tools are starting to develop into a management layer.

Predictions for the next 12 months

This summary has identified infrastructure provisioning and application packaging or configuration as essential tools to build the management layer for DevOps. I would like to conclude with a few predictions:

- All companies in the immutable infrastructure space will have tools to provision infrastructure, package applications, schedule applications and monitor applications.
- Configuration management companies will get the majority of enterprise budgets. It will take the immutable infrastructure companies more than 12 months to make significant progress with enterprises.
- AWS will grow by over 50 per cent and soak up a larger percentage of overall infrastructure budgets. The AWS tooling ecosystem is significantly more mature than that of its competitors, and more companies are moving away from on-premises data centres.

How to choose a DevOps management layer

The purpose of this article is to shed light on the crowded and opaque DevOps landscape. However, the obvious next question is, “How do I choose a DevOps management layer?” This question has many moving parts, and the best we can do is try to simplify some of the choices by answering a few questions:

- Are two or more individuals working on your team’s infrastructure (part-time contributors are counted)?
- If not, the two best options are either Amazon Web Services or Heroku. If no one has ops experience, Heroku is the best choice. If someone does have ops experience, AWS is the most full-featured option.
- If your answer to the above question is ‘Yes’, your team needs a way to manage the complexity of collaborating on infrastructure. Move on to the next question.
- Do you want to do build-time or runtime configuration?
 - If you want to do runtime configuration, use a configuration management tool such as Ansible, Puppet, Chef or Salt. If you want agent based configuration management, use Puppet or Chef. If you want ssh based configuration management, use Ansible or Salt.
 - If you want to do build-time configuration, which means building images and immutable infrastructure, use Atlas by HashiCorp, Mesosphere, Tectonic by CoreOS, or Docker Hub. Move on to the next question to select between these options.
- Do you want to automate the infrastructure provisioning process?
 - If you do, Atlas by HashiCorp is the only option. Atlas uses Terraform to lay out infrastructure as code and automate the provisioning process. Docker does have Docker Machine to provision Docker hosts; however, it is currently a command-line tool that only manages hosts, not a complete infrastructure.
 - If you do not want to automate the infrastructure provisioning process, any of the options by HashiCorp, Mesosphere, Docker, and CoreOS can be considered. However, it is difficult to deploy applications without a means to automate and reliably provision infrastructure.

This basic decision tree is by no means exhaustive, but should help in the initial decision-making process for selecting a DevOps management tool.



Author note: As an actor in the DevOps ecosystem, I certainly have biases. This summary is my view of the current DevOps environment backed by my experiences and the experiences of our customers. It is not exhaustive.



Note: This article has been taken from The New Stack and more such articles can be found at <http://thenewstack.io/>



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